Digit Queries

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2431 | Introductory Problems | https://cses.fi/problemset/task/2431

Consider an infinite string that consists of all positive integers in increasing order:

12345678910111213141516171819202122232425...

Your task is to process q queries of the form: what is the digit at position k in the string?

Input

The first input line has an integer q: the number of queries.

After this, there are q lines that describe the queries. Each line has an integer k: a 1-indexed position in the string.

Output

For each query, print the corresponding digit.

Constraints

- $1 \le q \le 1000$
- $1 \le k \le 10^{18}$

Example

Input
3 7 19 12

Output
7 4 1